New L J Institute of Engineering & Technology

SEM: 1st – CSE/CSE (AIML)/CSE(DS)

Sub: Mathematics-I

Sub Code : BE01R00041

Module 2: Basic Calculus

Weightage: 20%

Topics:

- **Chap:1 (5%)**
  Evaluation of improper integrals of Type-I and Type-II,
- **Chap: 2 (5%)**
  Beta and Gamma functions and their properties
- **Chap: 3 (10%)**
  Applications of definite integrals to evaluate surface areas and volumes of revolutions

❖ Chap 1: Improper Integral

▪ Definition: Improper Integrals

The integral $\int_a^b f(x)dx$ is called an improper integral if

(1) One or both limits of integration are infinite.

(2) Function $f(x)$ becomes infinite at a point within or at the end points of the interval of integration.

→ Improper integrals are classified into two kinds.

- ➢ Improper Integrals of First Kind (Type-I)
- ➢ Improper Integrals of Second Kind (Type-II)

▪ Definition: Improper Integrals of First Kind (Type-I)

It is a definite integral in which one or both limits of integration are infinite.

These integrals are evaluated as follows:

(1) If $f(x)$ is continuous on $[a, \infty)$ then

$$\int_a^{\infty} f(x)dx = \lim_{b\to\infty} \int_a^b f(x)dx$$

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(2) If $f(x)$ is continuous on $(-\infty, b]$ then

$$\int_{-\infty}^{b} f(x)dx = \lim_{a\to-\infty}\int_{a}^{b} f(x)dx$$

(3) If $f(x)$ is continuous on $(-\infty, \infty)$ then

$$\int_{-\infty}^{\infty} f(x)dx = \lim_{a\to-\infty}\int_{a}^{0} f(x)dx + \lim_{b\to\infty}\int_{0}^{b} f(x)dx$$

- The improper integral is said to converge (or exist) when the limit in RHS of (1),(2) and (3) exist (or finite);Otherwise, it is said to diverge.
- **For Examples**

(1) $\int_1^\infty \frac{1}{x^2}\, dx$

→The upper limit of integration is infinite

(2) $\int_{-\infty}^0 x \sin x\, dx$

→The lower limit of integration is infinite

(3) $\int_{-\infty}^\infty \frac{dx}{1+x^2}$

→ Both limits of integration are infinite.

Example: 1 Evaluate $\int_1^\infty \frac{1}{\sqrt{x}}\, dx$

Solution:

$$\int_1^\infty \frac{1}{\sqrt{x}}\, dx = \lim_{b\to\infty}\int_1^b \frac{1}{\sqrt{x}}\, dx$$

$$= \lim_{b\to\infty} (2\sqrt{x})_1^b$$

$$= \lim_{b\to\infty} (2\sqrt{b} - 2)$$

$= \infty$; Infinite

Example: 2 Evaluate the improper integral $\int_1^\infty \frac{1}{x^2} dx$

Solution:

$$\int_1^\infty \frac{1}{x^2}\, dx = \lim_{b\to\infty}\int_1^b \frac{1}{x^2}\, dx$$

$$= \lim_{b\to\infty} \left|\frac{-1}{x}\right|_1^b$$

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$$= \lim_{b\to\infty}\left(-\frac{1}{b}+1\right)$$

$$= 1 \text{ ; finite}$$

Example: 3 Evaluate $\int_0^\infty \frac{dx}{x^2+1}$

Solution:

$$\int_0^\infty \frac{dx}{x^2+1} = \lim_{b\to\infty}\int_0^b \frac{dx}{x^2+1}$$

$$= \lim_{b\to\infty}(\tan^{-1}x)_0^b$$

$$= \lim_{b\to\infty}(\tan^{-1}b - \tan^{-1}0)$$

$$= \lim_{b\to\infty}\tan^{-1}b$$

$$= \tan^{-1}\infty$$

$$= \frac{\pi}{2} \text{ ; finite}$$

Example: 4 Evaluate $\int_1^\infty \frac{1}{1+x^2}dx$

Solution:

$$\int_1^\infty \frac{1}{x^2+1}\,dx = \lim_{b\to\infty}\int_1^b \frac{1}{x^2+1}\,dx$$

$$= \lim_{b\to\infty}|\tan^{-1}x|_1^b$$

$$= \lim_{b\to\infty}(\tan^{-1}b - \tan^{-1}1)$$

$$= \tan^{-1}(\infty) - \frac{\pi}{4}$$

$$= \frac{\pi}{2} - \frac{\pi}{4}$$

$$= \frac{\pi}{4} \text{ ; finite}$$

Example: 5 Evaluate $\int_0^\infty \frac{dx}{a^2+x^2}; a > 0$

Solution:

$$\int_0^\infty \frac{dx}{a^2+x^2} = \lim_{b\to\infty}\int_0^b \frac{dx}{a^2+x^2}$$

$$= \lim_{b\to\infty}\left|\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)\right|_0^b$$

$$= \left(\frac{1}{a}\right)\lim_{b\to\infty}\left[\tan^{-1}\left(\frac{b}{a}\right) - 0\right]$$

$$= \left(\frac{1}{a}\right)\tan^{-1}(\infty) = \frac{\pi}{2a} \text{ ; finite}$$

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Example: 6 Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\, dx$.

Solution:

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\, dx = \lim_{a\to-\infty} \int_{a}^{0} \frac{1}{1+x^2}\, dx + \lim_{b\to\infty} \int_{0}^{b} \frac{1}{1+x^2}\, dx$$

$$= \lim_{a\to-\infty} |\tan^{-1} x|_a^0 + \lim_{b\to\infty} |\tan^{-1} x|_0^b$$

$$= \lim_{a\to-\infty} (0 - \tan^{-1} a) + \lim_{b\to\infty} (\tan^{-1} b - 0)$$

$$= \frac{\pi}{2} + \frac{\pi}{2}$$

$$= \pi; \text{ finite}$$

Example: 7 Evaluate $\int_{0}^{\infty} \frac{dv}{(1+v^2)\,(1+\tan^{-1} v)}$.

Solution:

$$\int_{0}^{\infty} \frac{dv}{(1+v^2)\,(1+\tan^{-1} v)} = \lim_{b\to\infty} \int_{0}^{b} \frac{\frac{1}{(1+v^2)}}{(1+\tan^{-1} v)}\, dv$$

$$= \lim_{b\to\infty} |\log|1+\tan^{-1} v||_0^b$$

$$= \lim_{b\to\infty} [\log|1+\tan^{-1} b| - \log 1]$$

$$= \lim_{b\to\infty} log(1+tan^{-1}\infty) - 0$$

$$= \log\left(1+\frac{\pi}{2}\right); \text{ finite}$$

Example: 8 Evaluate $\int_{-\infty}^{0} e^{2x}\, dx$.

Solution:

$$\int_{-\infty}^{0} e^{2x}\, dx = \lim_{a\to-\infty} \int_{a}^{0} e^{2x}\, dx$$

$$= \lim_{a\to-\infty} \left|\frac{e^{2x}}{2}\right|_a^0$$

$$= \lim_{a\to-\infty} \left(\frac{1}{2} - \frac{1}{2}\, e^{2a}\right)$$

$$= \frac{1}{2} - 0$$

$$= \frac{1}{2}\ ; \text{ finite}$$

Example: 9 Evaluate the improper integral $\int_{5}^{\infty} \frac{5x}{(1+x^2)^3} dx$ **OR**

Investigate convergence of the following integrals: $\int_{5}^{\infty} \frac{5x}{(1+x^2)^3} dx$

Solution:

$$\int_{5}^{\infty} \frac{5x}{(1+x^2)^3} dx = \lim_{b\to\infty} \int_{5}^{b} \frac{5x}{(1+x^2)^3} dx$$

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$Take\ 1 + x^2 = t \Rightarrow 2xdx = dt \Rightarrow xdx = \frac{dt}{2}$

Limit: $When\ x = 5 \Rightarrow t = 26\ and\ x = b \Rightarrow t = 1 + b^2$

$$I = \lim_{b\to\infty} \int_{t=26}^{1+b^2} \frac{5dt}{2(t)^3}$$

$$= \frac{5}{2} \lim_{b\to\infty} \left|\frac{t^{-2}}{-2}\right|_{26}^{1+b^2}$$

$$= \frac{5}{2704};\ finite$$

Hence, the integrand is convergent.

Example: 10 Evaluate $\int_{-\infty}^{\infty} \frac{1}{e^x+e^{-x}}\, dx$ by using improper integral.

Solution:

$$\int_{-\infty}^{\infty} \frac{1}{e^x + e^{-x}} dx = \int_{-\infty}^{\infty} \frac{e^x}{e^{2x} + 1} dx$$

$$= \int_{-\infty}^{0} \frac{e^x}{e^{2x}+1} dx + \int_{0}^{\infty} \frac{e^x}{e^{2x}+1} dx$$

$$= \lim_{a\to-\infty} \int_{a}^{0} \frac{e^x}{e^{2x}+1} dx + \lim_{b\to\infty} \int_{0}^{b} \frac{e^x}{e^{2x}+1} dx$$

$\boldsymbol{Take\ \ e^x = t \Longrightarrow e^x dx = dt}$

$\Longrightarrow Limit\ for\ first\ integral$:

$When\ x = a \Longrightarrow t = e^a\ and\ \ x = 0 \Longrightarrow t = \ e^0 = 1$

$\Longrightarrow Limit\ for\ second\ integral$:

$When\ x = 0 \Longrightarrow t = \ e^0 = 1\ and\ \ x = b \Longrightarrow t = \ e^b$

$$I = \lim_{a\to-\infty} \int_{e^a}^{1} \frac{dt}{t^2 + 1} + \lim_{b\to\infty} \int_{1}^{e^b} \frac{dt}{t^2 + 1}$$

$$= \lim_{a\to-\infty} |\tan^{-1} x|_{e^a}^{1} + \lim_{b\to\infty} |\tan^{-1} x|_{1}^{e^b}$$

$$= \lim_{a\to-\infty} (\tan^{-1} 1 - \tan^{-1}\ e^a) + \lim_{b\to\infty} (\tan^{-1}\ e^b - \tan^{-1} 1)$$

$$= \left(\frac{\pi}{4} - 0\right) + \left(\frac{\pi}{2} - \frac{\pi}{4}\right) = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}\ \ ;\ \text{Finite}$$

$$\left[\begin{array}{c} \because \tan^{-1}\ e^{-\infty} = \tan^{-1}\left(\frac{1}{\infty}\right) = \tan^{-1}(0) = 0 \\ and\ \tan^{-1}\ e^{\infty} = \tan^{-1}(\infty) = \frac{\pi}{2} \end{array}\right]$$

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Example: 11 Show that $\int_2^\infty \frac{5}{e^x-6}dx$ converges.

Solution:

$$\int_2^\infty \frac{5}{e^x-6}dx = \lim_{b\to\infty}\int_2^b \frac{5}{e^x-6}dx$$

$\Rightarrow$ ***Take*** $e^x - 6 = t$

$\Rightarrow e^x dx = dt$

$\Rightarrow dx = \frac{dt}{t+6}$

Limit: $When\ x = 2 \Rightarrow t = e^2 - 6\ and$

$$x = b \Rightarrow t = e^b - 6$$

$$= \lim_{b\to\infty}\int_{e^2-6}^{e^b-6} \frac{5}{t}\left(\frac{dt}{t+6}\right)$$

$$= \lim_{b\to\infty}\int_{e^2-6}^{e^b-6} \frac{5}{6}\left(\frac{1}{t}-\frac{1}{t+6}\right)dt$$

$$= \lim_{b\to\infty}\frac{5}{6}\left[\log\frac{t}{t+6}\right]_{e^2-6}^{e^b-6}$$

$$= \left(\frac{5}{6}\right)\log\left(\frac{e^2}{e^2-6}\right)\ ;\ finite$$

Hence, the integrand is convergent.

Example: 12 Evalaute $\int_{-1}^\infty \frac{dt}{t^2+5t+6}$

Solution:

$$\int_{-1}^\infty \frac{dt}{t^2+5t+6} = \lim_{b\to\infty}\int_{-1}^b \frac{dt}{t^2+5t+6}$$

$\Rightarrow$ ***Now*** $\frac{1}{t^2+5t+6} = \frac{1}{(t+2)(t+3)} = \frac{A}{(t+2)} + \frac{B}{(t+3)}$

$\Rightarrow 1 = A(t+3) + B(t+2)$

$\Rightarrow Take\ t = -2\ \Rightarrow 1 = A$

$\Rightarrow Take\ t = -3\ \Rightarrow 1 = -B \Rightarrow B = -1$

$\Rightarrow \frac{1}{(t+2)(t+3)} = \frac{1}{(t+2)} - \frac{1}{(t+3)}$

$$\Rightarrow \int_{-1}^\infty \frac{dt}{t^2+5t+6} = \lim_{b\to\infty}\int_1^b \left[\frac{1}{(t+2)} - \frac{1}{(t+3)}\right]dt$$

$$= \lim_{b\to\infty}[ln(t+2) - ln(t+3)]_{-1}^b$$

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$$= \lim_{b\to\infty}\left[ln\left(\frac{t+2}{t+3}\right)\right]_{-1}^{b}$$

$$= \lim_{b\to\infty} ln\left(\frac{b+2}{b+3}\right) - ln\left(\frac{1}{2}\right)$$

$$= ln1 + ln2$$

$$= ln2;\ finite$$

Hence, the integrand is convergent.

Example: 13 Prove that $\int_1^\infty \frac{1}{x^p}dx$, converges when p> 1 and diverges when p≤ 1.

OR Determine the convergence of $\int_1^\infty \frac{dx}{x^p}$

Solution:

$$\int_1^\infty \frac{dx}{x^p} = \lim_{b\to\infty}\int_1^\infty \frac{1}{x^p}\,dx$$

$$= \lim_{b\to\infty}\left[\frac{x^{-p+1}}{-p+1}\right]_1^b, p \neq 1$$

Case: 1 $When\ p > 1$

$$\int_1^\infty \frac{dx}{x^p} = \lim_{b\to\infty}\left[\frac{b^{-p+1}-1}{-p+1}\right]_1^b$$

$$= \left[\frac{\frac{1}{\infty}-1}{1-p}\right]$$

$$= \frac{1}{1-p}$$, finite and the integral converges.

Case: 2 $When\ p < 1$

$$\int_1^\infty \frac{dx}{x^p} = \lim_{b\to\infty}\left[\frac{b^{-p+1}-1}{-p+1}\right]_1^b$$

$$= \left[\frac{\infty-1}{1-p}\right]$$

$= \infty$, and the integral diverges.

Case: 3 $When\ p = 1$

$$\int_1^\infty \frac{dx}{x^p} = \lim_{b\to\infty}\int_1^\infty \frac{1}{x}\,dx$$

$$= \lim_{b\to\infty}[\log x]_1^b$$

$= [\log\infty - \log 1] = \infty$,; $Infinite$ and the integral diverges.

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▪ Definition: Improper Integrals of Second Kind (Type-II)

It is a definite integral in which integrand become infinite (or unbounded or discontinuous) at one or more points within or at the end points of the interval of integration.

These integrals are evaluated as follows:

(1) If $f(x)$ is unbounded at $x = a$ then

$$\int_a^b f(x)dx = \lim_{c\to a}\int_c^b f(x)dx$$

(2) If $f(x)$ is unbounded at $x = b$ then

$$\int_a^b f(x)dx = \lim_{c\to b}\int_a^c f(x)dx$$

(3) If $f(x)$ is unbounded at $x = c,\ a < c < b$

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx = \lim_{c_1\to c}\int_a^{c_1} f(x)dx + \lim_{c_2\to c}\int_{c_2}^b f(x)dx$$

The improper integral is said to converge (or exist) when the limit in RHS of (1), (2) and (3) exist (or finite). Otherwise, it is said to diverge.

Example: 1 Evaluate $\int_0^3 \frac{1}{\sqrt{3-x}}\,dx$

Solution: The integrand $\frac{1}{\sqrt{3-x}}$ is unbounded at $x = 3$.

$$\int_0^3 \frac{1}{\sqrt{3-x}}\,dx = \lim_{c\to 3}\int_0^c \frac{1}{\sqrt{3-x}}\,dx$$

$$= \lim_{c\to 3}\left(-2\sqrt{3-x}\right)_0^c$$

$$= \lim_{c\to 3}\left(-2\sqrt{3-c} + 2\sqrt{3}\right)$$

$$= 2\sqrt{3}\ ;\ \text{(finite)}$$

Example: 2 Evaluate $\int_0^1 \frac{1}{1-x}\,dx$

Solution: The integrand $\frac{1}{1-x}$ is unbounded at $x = 1$.

$$\int_0^1 \frac{1}{1-x}\,dx = \lim_{c\to 1}\int_0^1 \frac{1}{1-x}\,dx$$

$$= \lim_{c\to 1}\left[-\log\{1-x\}\right]_0^c$$

$$= \lim_{c\to 1}\left[-\log(1-c) + log1\right]$$

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$=\infty$; (Infinite)

Example: 3 check the convergence of $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$

Solution: The integrand $\frac{1}{\sqrt{9-x^2}}$ is unbounded at $x = 3$

$$\int_0^3 \frac{dx}{\sqrt{9-x^2}} = \lim_{c\to 3} \int_0^c \frac{dx}{\sqrt{9-x^2}}$$

$$= \lim_{c\to 3} \left(\sin^{-1}\frac{x}{3}\right)_0^c$$

$$= \lim_{c\to 3} \left(\sin^{-1}\frac{c}{3} - \sin^{-1} 0\right)$$

$$= \sin^{-1} 1 - \sin^{-1} 0$$

$$= \frac{\pi}{2} \text{ ; (finite)}$$

Hence, the improper integral is cgt.

Example: 4 Evaluate $\int_0^3 \frac{dx}{(x-1)^{\frac{2}{3}}}$

Solution: The integrand $\frac{1}{(x-1)^{\frac{2}{3}}}$ is unbounded at $x = 1$.

$$\int_0^3 \frac{dx}{(x-1)^{\frac{2}{3}}} = \lim_{c_1\to 1} \int_0^{c_1} \frac{dx}{(x-1)^{\frac{2}{3}}} + \lim_{c_2\to 1} \int_{c_2}^3 \frac{dx}{(x-1)^{\frac{2}{3}}}$$

$$= \lim_{c_1\to 1} \left(\frac{(x-1)^{\frac{1}{3}}}{\frac{1}{3}}\right)_0^{c_1} + \lim_{c_2\to 1} \left(\frac{(x-1)^{\frac{1}{3}}}{\frac{1}{3}}\right)_{c_2}^{3}$$

$$= \lim_{c_1\to 1} 3\left[(c_1-1)^{\frac{1}{3}} - (-1)^{\frac{1}{3}}\right] + \lim_{c_2\to 1} 3\left[(3-1)^{\frac{1}{3}} - (c_2-1)^{\frac{1}{3}}\right]$$

$$= 3\left[0 - (-1)^{\frac{1}{3}}\right] + 3\left[2^{\frac{1}{3}} - 0\right]$$

$$= 3\left[2^{\frac{1}{3}} + 1\right] \text{ ; (finite)}$$

Example: 5 Evaluate $\int_0^3 \frac{dx}{x-1}$ if possible

Solution: The integrand is unbounded at $x = 1$

$\therefore$ Second Kind Improper Integral

$$\int_0^3 \frac{dx}{x-1} = \int_0^1 \frac{dx}{x-1} + \int_1^3 \frac{dx}{x-1}$$

$$= \lim_{c\to 1} \int_0^c \frac{dx}{x-1} + \lim_{c\to 1} \int_c^3 \frac{dx}{x-1}$$

$$= \lim_{c\to 1} |\log|x-1||_{x=0}^{c} + \lim_{c\to 1} |\log|x-1||_{x=c}^{3}$$

$$= \lim_{c\to 1}[\log|c-1| - 0] + \lim_{c\to 1}[\log|2| - \log|c-1|]$$

$$= (-\infty) \text{ ; (Infinite)}$$

Hence, the integrand is divergent.

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Example: 6 Evaluate the improper integral $\int_0^1 \frac{1}{\sqrt{x}} dx$

Solution: The integrand $\frac{1}{\sqrt{x}}$ is unbounded at $x = 0$

∴ Second Kind Improper Integral

$$\int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{c\to 0} \int_c^1 \frac{1}{\sqrt{x}}\, dx$$

$$= \lim_{c\to 0} \left|2\sqrt{x}\right|_{x=c}^{1}$$

$$= \lim_{c\to 0} \left(2 - 2\sqrt{c}\right)$$

$$= 2 \text{ ; (finite)}$$

Example: 7 Can we solve the integral $\int_0^5 \frac{1}{(x-2)^2} dx$ directly? Give the reason.

Solution: We cannot solve the integral dx directly as $x = 2$ is a point of discontinuity

, which lies in the interval $[0,5]$

it is Second Kind Improper Integral.

Example: 8 Evaluate $\int_0^\infty \frac{1}{x} dx$

Solution: The integrand $\frac{1}{\sqrt{x}}$ is unbounded at $x = 0$

∴ Second Kind Improper Integral

$$\int_0^\infty \frac{1}{x} dx = \lim_{b\to\infty} \int_0^b \frac{1}{x} dx$$

$$= \lim_{b\to\infty} |\log x|_{x=1}^{b}$$

$$= \lim_{b\to\infty} (\log x - 0) = \infty \text{ ; (Infinite)}$$

Hence, the integrand is divergent.

Example: 9 Evaluate $\int_0^5 \frac{1}{x^2}\, dx$.

Solution:

The integrand $\frac{1}{x^2}$ is unbounded at $x = 0$.

$$\int_0^5 \frac{1}{x^2}\, dx = \lim_{c\to 0} \int_c^5 \frac{1}{x^2}\, dx$$

$$= \lim_{c\to 0} \left(-\frac{1}{x}\right)_c^5$$

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$$= \lim_{c\to 0}\left[-\frac{1}{5}+\frac{1}{c}\right]$$

$$= -\frac{1}{5}+\infty$$

$$= \infty \text{ ; (Infinite)}$$

Example : 10 Check the convergence of $\int_0^1 \frac{dx}{1-x}$

Solution: The integrand $\frac{1}{1-x}$ is unbounded at $x = 1$

$$\int_0^1 \frac{dx}{1-x} = \lim_{c\to 1}\int_0^c \frac{dx}{1-x}$$

$$= \lim_{c\to 1} |-\log(1-x)|_0^c$$

$$= \lim_{c\to 1} [-\log(1-c)+\log 1]$$

$$= -\log(1-1)$$

$$= \infty \text{ ; (Infinite)}$$

Hence, the integrand is divergent.

Example: 11 Determine whether $\int_0^{\frac{\pi}{2}} \sec x dx$ converges or diverges

Solution: The integrand $\sec x$ is not continuous at $x = \frac{\pi}{2}$

∴ Second Kind Improper Integral

$$\int_0^{\frac{\pi}{2}} \sec x dx = \lim_{c\to\frac{\pi}{2}}\int_0^c \sec x dx$$

$$= \lim_{c\to\frac{\pi}{2}} |\log|\sec x+\tan x||_{x=0}^c$$

$$= \lim_{c\to\frac{\pi}{2}} \log|\sec c+\tan c|$$

$$= \log\left|\sec\frac{\pi}{2}+\tan\frac{\pi}{2}\right|$$

$$= \infty \text{ ; (Infinite)}$$

Hence, the integrand is divergent.

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Convergence and Divergence of Improper Integrals:

1. Direct Comparison Test

(a) If $f(x)$ and $g(x)$ are two continuous functions for $x \geq a$, such that $0 \leq f(x) \leq g(x)$, then

- $\int_a^\infty f(x)\ dx$ converges, if $\int_a^\infty g(x)\ dx$ converges and

  $\int_a^\infty f(x)\ dx \leq \int_a^\infty g(x)\ dx.$

(b) If $f(x)$ and $g(x)$ are two continuous functions for $x \geq a$, such that $f(x) \geq g(x)$, then

- $\int_a^\infty f(x)\ dx$ diverges, if $\int_a^\infty g(x)\ dx$ diverges.

2. Limit Comparison Test

If $f(x)$ and $g(x)$ are two continuous functions for $x \geq a$, such that $f(x) > 0, g(x) > 0$ and $\lim_{x\to\infty} \frac{f(x)}{g(x)} = l, 0 < l < \infty$ then

- $\int_a^\infty f(x)\ dx$ and $\int_a^\infty g(x)\ dx$ both converges or diverges together.

3. p- Integral Test

(a) $\int_a^\infty \frac{dx}{x^p}$; $a > 0$; converges when p> 1 and diverges when p≤ 1.

(b) $\int_0^a \frac{1}{x^p} dx$; $a > 0$; converges when p< 1 and diverges when p≥ 1.

Example: 1 Test the convergence of the improper integral $\int_1^\infty \frac{\cos x}{x^2}\ dx$.

Solution: $f(x) = \frac{\cos x}{x^2}$ and let $g(x) = \frac{1}{x^2}$

$$\frac{\cos x}{x^2} \leq \frac{1}{x^2} \text{ for } x \geq 1 \qquad [\because \cos x \leq 1]$$

$$\int_1^\infty g(x)\ dx = \int_1^\infty \frac{1}{x^2}\ dx$$

$$= \lim_{b\to\infty} \int_1^b \frac{1}{x^2}\ dx$$

$$= \lim_{b\to\infty} \left|-\frac{1}{x}\right|_1^b$$

$$= \lim_{b\to\infty} \left(-\frac{1}{b} + 1\right)$$

$$= 1$$

Thus, $\int_1^\infty \frac{1}{x^2}\ dx$ is convergent.

By direct comparison test, $\int_1^\infty \frac{\cos x}{x^2}\ dx$ is convergent.

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Example: 2 Test the convergence of the improper integral $\int_1^\infty e^{-x^2}\ dx$.

Solution: $f(x) = e^{-x^2}$

Since $x \geq 1 \Rightarrow x^2 \geq x$

$\Rightarrow -x^2 \leq -x$

$\Rightarrow e^{-x^2} \leq e^{-x}$

Take $g(x) = e^{-x}$

$$\int_1^\infty g(x)dx = \int_1^\infty e^{-x}dx$$

$$= \lim_{b\to\infty} \int_1^b e^{-x}dx$$

$$= \lim_{b\to\infty} -(e^{-b} - e^{-1})$$

$$= -\left(0 - \frac{1}{e}\right)$$

$$= \frac{1}{e}\ , finite$$

$\therefore \int_1^\infty g(x)dx$ Converges

$\therefore$ By Direct Comparison test, $\int_1^\infty e^{-x^2}dx$ is also convergent

Example: 3 Is $\int_4^\infty \frac{\sin^2 x}{\sqrt{x}\ (x-1)}dx$ convergent?

Solution: $f(x) = \frac{\sin^2 x}{\sqrt{x}\ (x-1)}$

Take $g(x) = \frac{1}{x^{3/2}}$

$$\frac{\sin^2 x}{\sqrt{x}\ (x-1)} \leq \frac{1}{x^{3/2}}$$

$$\int_4^\infty g(x)dx = \int_4^\infty \left(\frac{1}{x^{3/2}}\right) dx$$

Using p-Integral test, $\int_4^\infty \left(\frac{1}{x^{3/2}}\right)\ dx$ is convergent as $p = \frac{3}{2} > 1$

By Direct Comparison test, $\int_4^\infty \frac{\sin^2 x}{\sqrt{x}\ (x-1)}dx$ is convergent.

Example: 4 Check the convergence of the integral $\int_1^\infty \frac{sin^2 x}{x^2}\ dx$

Solution: $f(x) = \frac{sin^2 x}{x^2}$

Take $g(x) = \frac{1}{x^2}$

$$\frac{sin^2 x}{x^2} \leq \frac{1}{x^2}\ for\ x \geq 1$$

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$$\int_1^\infty g(x)dx = \int_1^\infty \left(\frac{1}{x^2}\right) dx$$

Using p-Integral test, $\int_1^\infty \left(\frac{1}{x^2}\right) dx$ is convergent as $p = 2 > 1$

By Direct Comparison test,

$\int_1^\infty \frac{sin^2 x}{x^2} dx$ is convergent.

Example: 5 Check the convergence of $\int_0^3 \frac{\cos 3x}{x^{5/2}} dx$

Solution: $f(x) = \frac{\cos 3x}{x^{5/2}}$

Take $g(x) = \frac{1}{x^{5/2}}$

$$\frac{\cos 3x}{x^{5/2}} \leq \frac{1}{x^{5/2}}$$

$$\int_0^3 g(x)dx = \int_0^3 \left(\frac{1}{x^{5/2}}\right) dx$$

$\int_0^3 \left(\frac{1}{x^{5/2}}\right) dx$ is convergent using p-Integral test with $p = \frac{5}{2} > 1$

By Direct Comparison test, $\int_0^3 \frac{\cos 3x}{x^{5/2}} dx$ is convergent.

Example: 6 Prove that $\int_1^\infty \frac{\cos x}{x^2} dx$ converges. (Dec-2015-OLD) [NLJIET]

Solution: $f(x) = \frac{\cos x}{x^2}$

Take $g(x) = \frac{1}{x^2}$

$$\frac{\cos x}{x^2} \leq \frac{1}{x^2} \; for \; x \geq 1$$

$$\int_1^\infty g(x)dx = \int_1^\infty \left(\frac{1}{x^2}\right) dx$$

Using p-Integral test, $\int_1^\infty \left(\frac{1}{x^2}\right) dx$ is convergent as $p = 2 > 1$

By Direct Comparison test, $\int_1^\infty \frac{\cos x}{x^2} dx$ is convergent

Example: 7 Test the convergence of the improper integral $\int_0^1 \frac{1}{x^2 + \sqrt{x}} dx$.

Solution: $f(x) = \frac{1}{x^2 + \sqrt{x}}$ and let $g(x) = \frac{1}{\sqrt{x}}$

$\frac{1}{x^2 + \sqrt{x}} < \frac{1}{\sqrt{x}}$; $0 < x \leq 1$; $[\because x^2 + \sqrt{x} > \sqrt{x}\,]$

$\int_0^1 g(x)\; dx = \int_0^1 \frac{1}{\sqrt{x}}\; dx$

$= \lim_{c \to 0} \int_c^1 \frac{1}{\sqrt{x}}\; dx$

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$$= \lim_{c\to 0} \left|2\sqrt{x}\right|_c^1$$

$$= \lim_{c\to 0} (2 - 2\sqrt{c})$$

$$= 2$$

Thus, $\int_0^1 \frac{1}{\sqrt{x}}\, dx$ is convergent.

By direct comparison test, $\int_0^1 \frac{1}{x^2+\sqrt{x}}\, dx$ is convergent.

Example: 8 Test the convergence of the improper integral $\int_4^\infty \frac{3x+5}{x^4+7}\, dx$.

Solution:

$f(x) = \frac{3x+5}{x^4+7}$ and let $g(x) = \frac{1}{x^3}$

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{\frac{3x+5}{x^4+7}}{\frac{1}{x^3}}$$

$$= \lim_{x\to\infty} \frac{3x^4+5x^3}{x^4+7}$$

$$= \lim_{x\to\infty} \frac{3+\frac{5}{x}}{1+\frac{7}{x^4}}$$

$$= 3$$

$$\int_4^\infty g(x)\, dx = \int_4^\infty \frac{1}{x^3}\, dx$$

$$= \lim_{b\to\infty} \int_4^b \frac{1}{x^3}\, dx$$

$$= \lim_{b\to\infty} \left|-\frac{1}{2x^2}\right|_4^b$$

$$= 0 + \frac{1}{2\,(4^2)}$$

$$= \frac{1}{32}$$

Thus, $\int_4^\infty \frac{1}{x^3}\, dx$ is convergent.

By limit comparison test, $\int_4^\infty \frac{3x+5}{x^4+7}\, dx$ is convergent.

Example: 9 Check the convergence of an improper integral $\int_5^\infty \frac{7x+4}{x^2+9}\, dx$ using comparison test.

Solution: $f(x) = \frac{7x+4}{x^2+9}$

Take $g(x) = \frac{1}{x}$

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{7x+4}{x^2+9} \times \frac{x}{1}$$

$$= 7, finite$$

$\therefore \int_5^\infty f(x)dx$ and $\int_5^\infty g(x)dx$ both converge or diverge together.

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$$\int_5^\infty g(x)dx = \int_5^\infty \left(\frac{1}{x}\right) dx$$

$$= \lim_{b\to\infty} \int_5^b \left(\frac{1}{x}\right) dx$$

$$= \lim_{b\to\infty} [\log b - \log 5]$$

$$\to \infty$$

$\int_5^\infty \left(\frac{1}{x}\right) dx$ is divergent

By Limit Comparison test, $\int_5^\infty \frac{7x+4}{x^2+9} dx$ is divergent.

Example: 10 Test the convergence of the improper integral $\int_0^1 \frac{1-e^{-x}}{x^3} dx$.

Solution: $f(x) = \frac{1-e^{-x}}{x^3}$ and let $g(x) = \frac{1}{x^3}$

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{\frac{1-e^{-x}}{x^3}}{\frac{1}{x^3}}$$

$$= \lim_{x\to\infty} (1 - e^{-x})$$

$$= 1$$

$$\int_0^1 g(x)\ dx = \int_0^1 \frac{1}{x^3}\ dx$$

$$= \lim_{c\to 0} \int_c^1 \frac{1}{x^3}\ dx$$

$$= \lim_{c\to 0} \left| -\frac{1}{2x^2} \right|_c^1$$

$$= \lim_{c\to 0} \left(-\frac{1}{2} + \frac{1}{2c^2} \right)$$

$$= \infty$$

Thus, $\int_0^1 \frac{1}{x^3}\ dx$ is divergent.

By limit comparison test, $\int_0^1 \frac{1-e^{-x}}{x^3}\ dx$ is divergent.

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